Comparative Evaluation of Supervised Learning Algorithms for Cardiovascular Disease Classification

MSc Project Dissertation

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University of Liverpool

Siddique Sayed

Supervisor: Karteek Sreenivasaiah

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Finally, I appreciate the experience I gained through this project. It gave me the chance to bring together what I had learned during the MSc and apply it to one complete piece of work, from cleaning the data and training the models to evaluating the results and reflecting on what I would do differently next time.

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Abstract

This dissertation presents a controlled comparison of seven supervised learning algorithms for cardiovascular disease classification. The study uses the publicly available Cardiovascular Disease Dataset from Kaggle, which contains 70,000 labelled records. The project was treated as an experiment: the same cleaned dataset, target definition, train-test split and evaluation measures were used to make the model comparison as consistent as possible.

The workflow covered data inspection, conservative data-quality filtering, feature engineering, feature-ablation analysis, preprocessing, stratified cross-validation, hyperparameter tuning and final evaluation on a held-out test set. The seven classifiers were Logistic Regression, Decision Tree, Random Forest, AdaBoost, Linear Support Vector Machine, K-Nearest Neighbours and Gaussian Naive Bayes. Performance was compared using Accuracy, Precision, Recall, F1 score and ROC-AUC rather than relying on one measure alone.

The tuned Random Forest achieved the highest Accuracy (0.7389) and ROC-AUC (0.8062), while tuned KNN achieved the highest Recall (0.6885) and F1 score (0.7197) in the primary comparison. The Decision Tree performed weakest and showed clear overfitting. Feature-ablation experiments also showed that adding BMI and pulse pressure did not materially improve predictive performance. Overall, the results show that the most suitable model depends on the evaluation objective and that a fair comparison requires attention to the full experimental pipeline, not only the final metric values.

Statement of Ethical Compliance

This project was conducted under Ethics Category B0. It used an existing publicly available dataset and involved no participant recruitment, direct participant activity or collection of new personal data. The work was carried out in accordance with the COMP702 ethical guidance.

Declaration of Generative AI Assistance

Generative AI tools, including ChatGPT and Claude, were used in a limited supporting role during the project. Their main use was to assist with coding, debugging and resolving technical issues, with some additional support for planning and clarification of concepts. The final code, analysis, results and written work were reviewed and checked against the project files and relevant source material before submission.

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1. Introduction & Background

1.1 Background and Motivation

Cardiovascular disease provides a clear binary classification setting with variables that are straightforward to interpret. The World Health Organization identifies raised blood pressure, raised blood glucose, raised blood lipids, excess body weight, tobacco use, harmful alcohol use and physical inactivity among important cardiovascular risk factors (World Health Organization, 2025). Several of these factors are represented directly in the Kaggle dataset, which made it suitable for comparing algorithms on a realistic but manageable tabular problem.

The project was therefore framed as a comparative computing experiment rather than a search for a deployable diagnostic model. This distinction shaped the work from the start. The question was not simply whether a classifier could achieve a high score, but whether different algorithm families behaved differently when they were given the same cleaned observations, the same target, the same held-out test set and the same set of evaluation measures. This emphasis is consistent with biomedical machine-learning guidance that stresses transparent reporting of data sources, validation procedures and model-selection decisions (Luo et al., 2016).

The study was therefore designed as a controlled comparison rather than a search for a single impressive score. Each algorithm was trained from the same cleaned observations and judged on the same held-out test set. Cross-validation was used during development so that tuning decisions were separated from the final evaluation. This design choice was important because apparently small differences in splitting, preprocessing or metric selection can change the ranking of models.

1.2 Problem Statement

The project addresses a binary supervised classification problem in which the target variable indicates the presence or absence of cardiovascular disease. The central question is how seven supervised learning algorithms compare when they are evaluated under the same experimental conditions, and what explains the differences that appear across Accuracy, Precision, Recall, F1 score and ROC-AUC.

1.3 Aim

The aim of this project is to comparatively evaluate selected supervised learning algorithms for cardiovascular disease classification using a consistent preprocessing, training, validation and evaluation framework.

1.4 Objectives

- Inspect and clean the cardiovascular disease dataset using reproducible data-quality rules.
- Explore the main relationships and distributions in the data.
- Investigate feature engineering and redundancy, including BMI and pulse pressure.
- Implement seven supervised classification algorithms representing different learning strategies.
- Use a consistent training, validation and test framework.
- Tune selected models using stratified cross-validation.
- Compare final models using Accuracy, Precision, Recall, F1 score and ROC-AUC.
- Interpret differences between models, including overfitting and Precision-Recall trade-offs.

1.5 Research Questions and Working Hypotheses

The research questions were tied directly to experiments. RQ1 was answered through the common held-out comparison across five metrics. RQ2 was examined by comparing default and tuned behaviour, particularly where large training-validation gaps suggested overfitting. RQ3 was tested through feature-ablation experiments that compared alternative feature representations under the same cross-validation procedure.

RQ1 asks how the seven algorithms differ in predictive performance under the same conditions. RQ2 asks whether hyperparameter tuning improves generalisation when a default model overfits or is sensitive to configuration. RQ3 asks whether engineered features such as BMI and pulse pressure add useful predictive information beyond the original measurements.

A working hypothesis for RQ1 was that ensemble methods would generalise more strongly than an unconstrained single Decision Tree because combining multiple learners can reduce variance. For RQ2, tuning was expected to reduce the gap between training and validation performance in models that initially overfit. For RQ3, engineered features were tested rather than assumed to be useful; their value was judged by cross-validated performance.

1.6 Related Work and Algorithm Background

Published cardiovascular prediction research does not identify one algorithm that is consistently optimal across datasets. Krittanawong et al. (2020) reported substantial variation in machine-learning performance across cardiovascular prediction tasks, while Bani Hani and Ahmad (2023) found considerable methodological diversity across ischemic heart-disease studies. A broader systematic review by Christodoulou et al. (2019) found no consistent performance advantage of machine-learning methods over Logistic Regression in low-risk-of-bias clinical prediction comparisons. This is particularly relevant to the present project because it supports the use of Logistic Regression as a serious benchmark rather than treating it as an outdated baseline. Together, these studies reinforce the need to describe preprocessing, validation and evaluation choices carefully before interpreting algorithm rankings.

Methodological reporting is also part of model validity. Luo et al. (2016) identify data leakage, overfitting, incomplete reporting of data sources and inadequate validation as recurring problems in biomedical predictive modelling. Their guidance supports the experimental emphasis adopted here: the training and final test roles are kept separate, preprocessing is included within the modelling pipeline, and final performance is reported using predefined measures rather than selecting a model from repeated inspection of the test data.

The present project differs from work that aims to build a deployable clinical decision system. Its main contribution is experimental: seven established algorithms are evaluated under a common pipeline, with explicit attention to cleaning, scaling, cross-validation, tuning, feature redundancy and multiple measures of predictive behaviour. This emphasis follows the principle that a comparative experiment is only interpretable when the procedure used to produce the results is described in sufficient detail.

1.6.1 Logistic Regression

Logistic Regression is a widely used linear classifier for binary outcomes. It models the log-odds of the positive class as a linear combination of the input variables and transforms the resulting score through the logistic function to obtain a probability between zero and one. A threshold,

commonly 0.5, can then convert the probability into a class prediction. Its strengths include computational efficiency, a relatively small number of parameters and interpretable coefficient directions. It also provides a useful baseline against which more flexible nonlinear models can be compared. A limitation is that the linear form may not capture complex interactions or nonlinear relationships unless additional transformations are introduced.

1.6.2 Decision Tree

Decision-tree induction has a long history in machine learning; Quinlan (1986) described recursive tree induction as a practical way of learning classification rules from examples. In this project, the method was useful not only as a classifier but also as a diagnostic comparison. Its strong training performance and much weaker held-out performance provided a clear example of the variance that can arise when a flexible tree is allowed to fit the training sample too closely.

A Decision Tree recursively partitions the feature space into regions that are increasingly homogeneous with respect to the target class. Trees are attractive because their decision structure can be interpreted directly, but an unrestricted tree can become highly specific to the training sample. This high variance can produce excellent training performance but weak performance on unseen data. In the present project the single Decision Tree provides both a nonlinear baseline and a useful contrast with Random Forest, which aggregates many trees to reduce the instability of a single learner.

1.6.3 Random Forest

Random Forest is an ensemble method introduced by Breiman (2001). It combines multiple tree predictors, typically using bootstrap samples of the training data and random subsets of features during splitting. Predictions are aggregated across trees. The use of many partially decorrelated trees can reduce variance relative to a single Decision Tree and often gives strong performance on structured tabular data. Random Forest also offers impurity-based feature importance, although such importance should be interpreted as model-specific predictive contribution rather than evidence of causality. In this project Random Forest was of particular interest because the default model showed overfitting, creating a clear opportunity to evaluate whether depth and leaf-size constraints could improve generalisation.

1.6.4 AdaBoost

AdaBoost is a boosting method derived from the work of Freund and Schapire (1997). Rather than fitting many learners independently, boosting constructs a sequence of weak learners and progressively places greater emphasis on observations that previous learners handled poorly. The final prediction combines the weak learners using weights. AdaBoost was included because it represents a different ensemble strategy from Random Forest: boosting focuses on sequential error correction, whereas Random Forest primarily reduces variance through aggregation and randomisation.

1.6.5 Linear Support Vector Machine

Support Vector Machines seek a separating decision boundary with a large margin between classes. The foundational support-vector formulation was described by Cortes and Vapnik (1995). The present project used a linear SVM implementation because the cleaned dataset contains more than sixty-eight thousand observations and repeated validation with a nonlinear kernel would increase computational cost substantially. The linear variant therefore provides a practical

margin-based comparison while preserving the core idea of maximising separation between classes. Scaling is important because the geometry of a margin-based classifier is influenced by the magnitude of the input variables.

1.6.6 K-Nearest Neighbours

K-Nearest Neighbours is a non-parametric, instance-based classifier. An unseen observation is assigned a class according to the labels of nearby training observations in feature space. The general nearest-neighbour principle has a long history in pattern recognition (Cover and Hart, 1967). KNN has very different behaviour from parametric models because it does not estimate a global set of coefficients. Its performance depends strongly on the distance metric, the number of neighbours and the scale of the features. This makes standardisation particularly important. KNN was included to determine whether local neighbourhood structure in the cardiovascular data could compete with linear, tree-based and ensemble approaches.

1.6.7 Gaussian Naive Bayes

Gaussian Naive Bayes is a probabilistic classifier based on Bayes' rule and a conditional-independence assumption between features given the class. Continuous features are commonly modelled using Gaussian class-conditional distributions. The independence assumption is unlikely to hold exactly in this dataset, particularly because several physiological variables are correlated. Nevertheless, Gaussian Naive Bayes provides a computationally efficient probabilistic baseline and helps show how a simple generative classifier behaves relative to more flexible discriminative models.

1.7 Evaluation Measures

Each metric was retained because it answers a different question. Accuracy asks how often the model is correct overall. Precision focuses on the reliability of positive predictions, while Recall focuses on how many true positive cases are recovered. F1 is useful when neither Precision nor Recall should dominate the comparison. ROC-AUC adds a threshold-independent view of ranking performance. Sokolova and Lapalme (2009) demonstrate that classification measures are not interchangeable, and Fawcett (2006) explains why ROC analysis is useful when comparing classifiers across operating thresholds.

This mattered in the final results. Random Forest and KNN were close in Accuracy, yet KNN produced a higher Recall and F1 whereas Random Forest produced stronger Precision and ROC-AUC. If only Accuracy had been reported, this difference in behaviour would have been hidden. The multi-metric design was therefore not an extra presentation choice; it was central to the research question.

1.8 Cross-Validation and Reproducibility

Five folds were selected as a practical compromise. The dataset is large enough that each validation fold still contains many observations, while five repeated fits per candidate keep the computational cost manageable when several algorithms and hyperparameter settings are compared. Stratification was retained because the target is binary and the intention was to keep the class proportions stable across folds. Although Kohavi (1995) reported strong results for ten-fold stratified cross-validation in his experiments, the present project used five folds to balance repeated validation with the cost of training seven classifiers on more than sixty-eight thousand cleaned records.

Cross-validation estimates model behaviour across repeated partitions of the available training data. In k-fold cross-validation, the training data are divided into k folds; each fold is used once for validation while the remaining folds are used for fitting. Stratified K-Fold preserves approximately the class proportions in each fold. Scikit-learn was used to implement this validation strategy as part of the modelling pipeline (Pedregosa et al., 2011). Five folds were used here as a balance between repeated validation and computational cost.

The project was implemented in Python using scikit-learn, a machine-learning library designed to provide consistent interfaces to a wide range of supervised and unsupervised methods (Pedregosa et al., 2011). Pipelines and column transformers were used to keep preprocessing coupled to model fitting. This is important because transformations should be fitted on training data rather than on the complete dataset, which would otherwise risk information leakage. This separation of preprocessing, validation and final testing is also consistent with biomedical predictive-modelling guidance that stresses the need to prevent leakage and report validation procedures clearly (Luo et al., 2016).

2. Design & Implementation

Because this project is research-focused, the design is mainly the design of the experiment. The aim was to build a pipeline that could be repeated and that would give each model a fair opportunity to learn from the same data. The implementation then turned that design into a set of Python notebooks, scikit-learn pipelines, saved model files and result tables used to document and compare the experimental outputs.

2.1 Dataset Source and Variables

The dataset was obtained from Svetlana Ulianova's "Cardiovascular Disease dataset" on Kaggle. The data card reports 70,000 records, eleven input features and one binary target (Ulianova, 2019). Age is recorded in days; height and weight are numerical; systolic and diastolic blood pressure are recorded as examination measurements; cholesterol and glucose use three ordered levels; and smoking, alcohol intake and physical activity are binary indicators. The mix of continuous, ordinal and binary variables made the dataset useful for comparing algorithms with different assumptions about scale and decision boundaries.

The sample size also allowed an 80/20 train-test split while leaving enough observations for five-fold cross-validation inside the training partition. This was useful for the project because model selection could be repeated without reducing the final test set to a very small sample.

Using a secondary public dataset also places limits on the study. The project did not control the original measurement process, and the Kaggle page provides a data dictionary but limited information about recruitment, measurement quality or sampling. The page also lists the licence as unknown. These issues do not prevent a methodological comparison, but they restrict how far the results can be generalised and are treated explicitly as limitations.

Variable	Description	Original type	Role
age	Age in days	Integer	Input
gender	Gender code	Categorical	Input
height	Height in cm	Integer	Input
weight	Weight in kg	Float	Input
ap_hi	Systolic blood pressure	Integer	Input
ap_lo	Diastolic blood pressure	Integer	Input
cholesterol	1 normal, 2 above normal, 3 well above normal	Ordinal	Input
gluc	1 normal, 2 above normal, 3 well above normal	Ordinal	Input
smoke	Smoking status	Binary	Input
alco	Alcohol intake	Binary	Input
active	Physical activity	Binary	Input
cardio	Presence/absence of CVD	Binary	Target

2.2 Initial Data Inspection

Initial inspection was deliberately separated from cleaning. The first checks covered shape, column names, data types, duplicate rows, missing values and target balance. No missing values were found, and the target was almost evenly divided between the two classes in the raw data. This meant that class imbalance was not a major issue, but stratification was still used so that the balance would be maintained after splitting.

The more important issue was plausibility. Several rows contained blood-pressure values that were difficult to interpret as ordinary measurements, including diastolic values reaching 1100, systolic values as low as 12 and negative pulse pressure after subtracting diastolic from systolic pressure. Extreme heights and weights were also present. Rather than correcting individual values manually, a reproducible set of conservative rules was applied to the full dataset.

Initial inspection confirmed 70,000 observations. No missing values were identified in the raw table, and the target was almost exactly balanced: 35,021 records in class 0 and 34,979 in class 1. After cleaning, 34,676 class-0 records and 33,951 class-1 records remained, corresponding to 50.53% and 49.47% of the cleaned dataset respectively. This balance makes Accuracy easier to interpret than it would be in a strongly imbalanced dataset, but it does not remove the need for Precision, Recall, F1 and ROC-AUC because similarly accurate classifiers can still distribute errors differently.

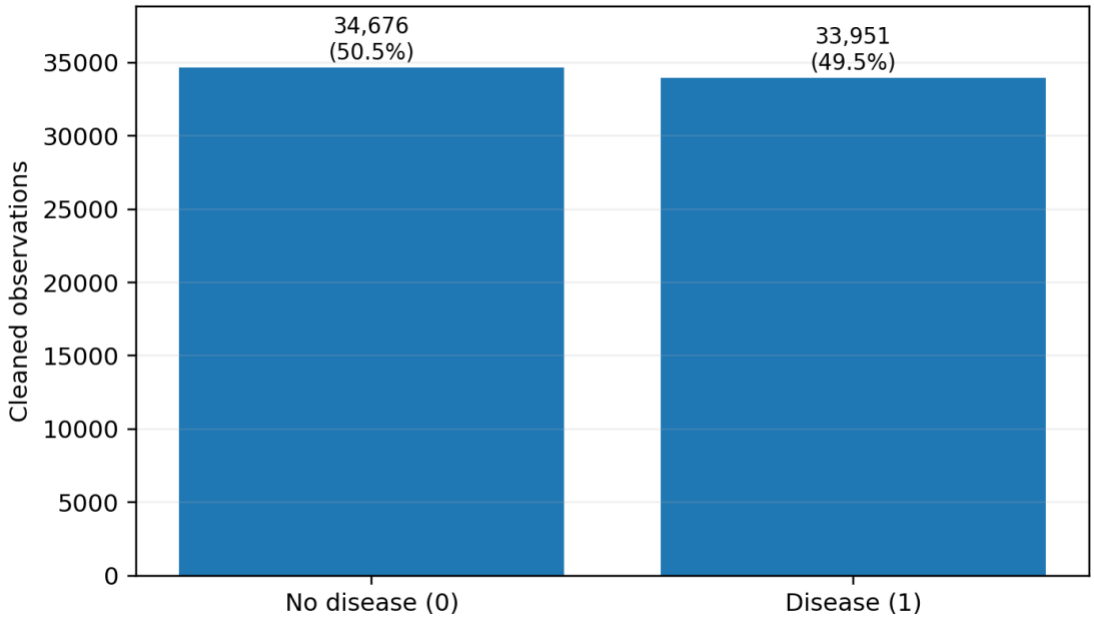


Figure 2.1. Target-class distribution after data cleaning.

The main data-quality issue was not missingness but implausible numerical values. Examples included extremely large diastolic measurements, very low systolic measurements, negative pulse pressure after subtraction, and unusually extreme height values. Such values can have a disproportionate effect on distance-based and linear methods and can create artificial split opportunities for trees. A reproducible filtering stage was therefore required before modelling.

2.3 Cleaning Decisions

The rules were implemented as filters so that the same logic could be rerun from the raw file. Height was restricted to 100-230 cm and weight to 30-250 kg. Systolic pressure was restricted to 70-300 and diastolic pressure to 40-200, with an additional requirement that systolic pressure exceed diastolic pressure. BMI was calculated after the height and weight checks and restricted to 10-70. These are described as plausibility rules rather than universal clinical boundaries because the project did not attempt to establish medical diagnostic thresholds.

The cleaning removed 1,373 observations and retained 68,627, or approximately 98.04% of the original dataset. Retaining such a large proportion was useful because it reduced the chance that the final comparison was driven mainly by aggressive data removal. At the same time, the removed rows contained exactly the type of extreme measurements that could distort scaling, distances and split thresholds if they were left untreated.

The identifier column was removed because it encodes row identity rather than a characteristic that should generalise to unseen observations. Age was converted from days to years using age/365.25. This does not add new information but makes the variable easier to interpret in descriptive analysis and model explanations.

2.4 Feature Engineering and Redundancy

Three transformations were explored. Age was converted from days to years because the conversion makes plots and descriptive statistics easier to interpret without changing the ordering of observations. BMI was derived from height and weight using weight divided by squared height in metres. Pulse pressure was derived as systolic minus diastolic blood pressure. These transformations were initially kept because each has an intuitive interpretation, but intuition alone was not treated as sufficient justification for keeping them in the predictive model.

The correlation and variance-inflation analysis made the redundancy visible. Pulse pressure is exactly determined by the two blood-pressure variables, so including all three creates a deterministic linear relationship. BMI similarly carries information already present in height and weight. The ablation experiment was therefore used to test whether replacing or extending the original variables with the engineered versions actually changed predictive performance. The result was essentially no change, which supported the simpler final representation.

The stronger issue was correlation among predictors. Systolic pressure correlated strongly with pulse pressure, while BMI correlated strongly with weight. Variance Inflation Factor analysis became extreme when mathematically related variables were included together: systolic pressure, diastolic pressure and pulse pressure produced effectively infinite VIF values because pulse pressure is an exact linear combination of the other two. Weight and BMI also produced very high VIF values when height, weight and BMI were combined. For this reason, the final modelling feature set avoided including all source and derived versions together.

Categorical association analysis using chi-square tests and Cramér's V showed that cholesterol had the strongest categorical association with cardio among the examined discrete variables, while glucose had a smaller association and gender, smoking, alcohol use and activity had weak effect sizes. Because the dataset is large, statistical significance alone was not treated as evidence of practical importance. Effect size and predictive performance were given greater weight.

The final modelling features were age_years, gender, BMI, ap_hi, ap_lo, cholesterol, gluc, smoke, alco and active. This ten-feature representation retained demographic, physiological, metabolic and lifestyle information while avoiding simultaneous inclusion of height, weight, BMI and pulse pressure. The choice was supported by subsequent ablation results, which showed that alternative representations using height and weight or adding pulse pressure produced almost identical Logistic Regression performance.

2.5 Experimental Pipeline

The order of operations was important. Cleaning and deterministic feature creation were completed before the train-test split because these steps used fixed rules rather than estimates learned from the target. In contrast, StandardScaler was fitted only inside the modelling pipeline on training folds. This prevents information about the mean and standard deviation of the validation or test observations from leaking into training. The same principle was followed during hyperparameter selection: validation folds were used to compare candidate configurations, while the held-out test set was reserved for the final comparison.

The final experimental sequence was kept deliberately simple: data inspection came first, followed by cleaning and feature engineering. Only after the final feature representation had been decided was the dataset split into training and test partitions. Model selection and tuning were then carried out using the training partition, while the held-out test set was reserved for the final comparison.

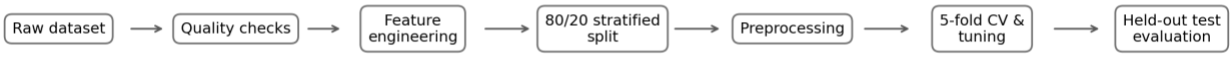


Figure 2.2. Experimental pipeline used for the supervised-learning comparison.

2.6 Train-Test Split

The cleaned data were divided into 80% training data and 20% test data using train_test_split with random_state=42 and stratify=y. The 80% training partition provided a large sample for fitting and five-fold cross-validation, while the 20% test partition remained large enough to support stable final metric estimates. Stratification preserved the class proportions across the two partitions. The fixed random state made the split reproducible so that differences between algorithm results were not caused by each model receiving a different random partition.

2.7 Preprocessing and Scaling

The numerical variables age_years, BMI, ap_hi and ap_lo were standardised using StandardScaler. Standardisation centres each selected feature using the training-set mean and scales it by the training-set standard deviation. This is especially relevant to KNN because distances depend directly on scale, and to Linear SVM and Logistic Regression because the optimisation and regularisation behaviour can be influenced by different feature magnitudes. The categorical or discrete variables gender, cholesterol, gluc, smoke, alco and active were passed through without scaling.

Tree-based models do not require feature scaling because split ordering is invariant to monotonic transformations of individual variables. Applying a common preprocessing structure therefore does not benefit Random Forest or Decision Tree in the same way that it benefits KNN or SVM. This is an important limitation of the idea of a “fully identical” pipeline: a fair comparison should hold the data and evaluation protocol constant, but algorithm-specific preprocessing can still be appropriate. The implementation used scikit-learn ColumnTransformer and Pipeline so that transformations were fitted on training data inside the modelling workflow rather than on the full dataset.

2.8 Algorithms and Cross-Validation

Seven classifiers were selected to represent different supervised learning strategies: Logistic Regression as a linear probabilistic baseline; Decision Tree as a single nonlinear tree; Random Forest as a bagging-style ensemble; AdaBoost as a boosting ensemble; Linear SVM as a margin-based classifier; KNN as a distance-based non-parametric model; and Gaussian Naive Bayes as a probabilistic generative baseline. This diversity was intentional. Comparing several algorithms from the same family would provide less information about how different inductive biases interact with the dataset.

Model evaluation during development used five-fold Stratified K-Fold cross-validation with shuffling and random_state=42. In each cross-validation iteration, four folds were used for fitting and one for validation, so every training observation contributed to validation exactly once per complete five-fold run. Stratification preserved approximately the target-class proportion in each fold. Five folds were selected as a practical compromise: more folds would increase computation for models such as Random Forest and KNN without necessarily changing the overall conclusions, while a single validation split would make model selection more dependent on one particular partition.

2.9 Hyperparameter Tuning

Tuning was targeted at models where the initial results suggested that configuration mattered. The default Decision Tree showed severe overfitting. The default Random Forest also had a large train-validation gap, despite performing better than the single tree, and default KNN showed a smaller but still meaningful gap. Rather than tuning every model simply because tuning was available, the project concentrated effort where there was evidence that complexity or neighbourhood size could affect generalisation.

For Random Forest, the selected tuned configuration used 300 estimators, maximum depth 10, minimum samples split 5, minimum samples leaf 8 and max_features="log2". These settings regularised the forest by limiting tree depth and preventing very small leaves. The result was a much smaller train-validation gap than the default forest. KNN was also tuned through cross-validation over its neighbourhood configuration and produced a substantial improvement over the default KNN. The comparison therefore uses the tuned KNN estimator associated with the final saved experimental results.

2.10 Development Environment and Realisation

Development was iterative rather than linear. Some early notebook runs failed because the dataframe had not been created in the current kernel, a missing optional package was imported, and the original age column had already been renamed to age_years. These were ordinary

implementation problems, but they exposed a weakness in relying on notebook state. The later notebooks therefore became more self-contained: they loaded the cleaned file explicitly, defined the feature lists near the top and saved important result tables to CSV.

The same lesson applied when results were transferred between notebooks and saved files. Important feature lists, preprocessing assumptions and result tables were therefore loaded explicitly rather than relying on notebook memory.

The project was implemented in Python, with Jupyter Notebook used for data inspection, preprocessing, statistical analysis and model development. Pandas and NumPy were used for data handling, while scikit-learn provided the preprocessing pipelines, classifiers, cross-validation tools and evaluation metrics. Model outputs and summary tables were saved so that the final comparison could be reproduced without manually copying results between notebooks.

The implementation changed as issues became visible. Early analysis exposed extreme blood-pressure and anthropometric values, so the cleaning stage was made explicit rather than treating the raw data as ready for modelling. A nonlinear SVC was also considered, but a Linear SVM was used because repeated cross-validation on nearly 70,000 observations made the linear version more practical. The default Decision Tree and Random Forest showed large training-validation gaps, which led to a stronger focus on regularisation and tuning.

The final implementation was organised around Jupyter notebooks for data inspection, preprocessing, statistical analysis, model development and evaluation. Important outputs were saved to CSV files so that later stages could use confirmed results consistently.

2.11 Fairness and Reproducibility

Fairness in this dissertation means procedural fairness between algorithms, not demographic fairness. Each final classifier was compared on the same test rows and using the same metric definitions. The split was created once and reused; the target definition did not change between models; and the final model table was generated from the same held-out set. This does not guarantee that every model received its individually optimal preprocessing or search budget, but it does make the reported comparison understandable and repeatable.

Several controls were used to make the comparison fair. Every classifier was evaluated on the same cleaned dataset, target definition, final feature set, training partition and held-out test partition. The same five headline metrics were reported for all models. Cross-validation was used for model selection and tuning rather than choosing configurations directly from test results. A fixed random state made stochastic splits reproducible. Scikit-learn pipelines kept preprocessing coupled to the model workflow. At the same time, the experiment acknowledges that “fair” does not mean every algorithm must receive identical preprocessing regardless of its mathematical requirements. Scaling is relevant to KNN, SVM and Logistic Regression but not required by tree learners. The final test partition was therefore treated as an estimate of generalisation after model-development decisions had been made, rather than as a resource for repeatedly choosing models. This distinction is central to avoiding optimistic performance estimates and is consistent with established guidance on predictive-model development (Luo et al., 2016; Pedregosa et al., 2011).

3. Testing & Evaluation

3.1 Testing the Data and Software Pipeline

Testing was carried out at several levels. At the data level, assertions and summary checks were used to confirm that cleaned values fell inside the intended ranges and that the target remained binary. At the modelling level, output shapes, class predictions and probability/decision scores were checked before metrics were calculated. Stored model outputs were compared with the benchmark CSV files so that later analysis did not silently change the confirmed results.

This distinction between software testing and model evaluation is important. Software testing asks whether the implementation behaves as intended: files load, preprocessing runs, models return outputs and charts display the expected data. Model evaluation asks a different question: how well the trained classifier generalises to unseen examples. The dissertation reports both, but the final Accuracy, Precision, Recall, F1 and ROC-AUC belong to the evaluation of predictive behaviour rather than to software correctness.

Testing took place throughout development rather than only at the end. Dataset checks confirmed the expected columns, target values and class counts before model fitting. The cleaned dataset was reloaded from the saved CSV to make sure later notebooks used the same data representation. The fixed random state and stratified split were also checked so that repeated runs produced the same partitions.

For the modelling pipeline, the main checks were that preprocessing was fitted using the training data, that every classifier received the same final feature set, and that the final metrics were calculated on the same held-out test observations. Saved model outputs were compared against the stored result tables to catch accidental differences between notebook runs. No human-participant software evaluation was carried out.

Check	Expected condition	Outcome
Cleaned dataset	68,627 records available	Pass
Target values	Binary values 0 and 1 only	Pass
Final feature set	All 10 modelling features present	Pass
Train/test split	80/20 stratified split reproducible with random_state=42	Pass
Saved model loading	Final model files load successfully	Pass
Stored results	Notebook/model outputs consistent with saved comparison tables	Pass

Table 3.1. Summary of reproducibility and data/model checks performed before final model evaluation.

3.2 Final Held-Out Model Comparison

Table 3.2 reports the final held-out test results for the seven classifiers. The Random Forest and KNN entries correspond to the tuned configurations used in the primary comparison. The results

demonstrate that the top models are relatively close on Accuracy, while their Precision, Recall and F1 profiles differ.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7305	0.7579	0.6689	0.7106	0.7943
Decision Tree	0.6417	0.6387	0.6349	0.6368	0.6418
Random Forest (Tuned)	0.7389	0.7698	0.6736	0.7185	0.8062
AdaBoost	0.7309	0.7699	0.6504	0.7051	0.7984
Linear SVM	0.7277	0.7605	0.6563	0.7046	0.7943
KNN (Tuned)	0.7347	0.7539	0.6885	0.7197	0.7970
Gaussian Naive Bayes	0.7111	0.7595	0.6087	0.6758	0.7816

Table 3.2. Final held-out test performance of the seven classifiers.

The tuned Random Forest achieved the highest Accuracy, 0.7389, and the highest ROC-AUC, 0.8062. KNN achieved the highest Recall, 0.6885, and the highest F1 score, 0.7197, although the F1 difference between KNN and Random Forest was small. AdaBoost produced the highest Precision, 0.7699, by a very small margin over Random Forest at 0.7698. This immediately demonstrates why one metric cannot define a universal winner. Random Forest was strongest on overall correctness and discrimination, KNN provided a more recall-oriented balance, and AdaBoost slightly prioritised precision.

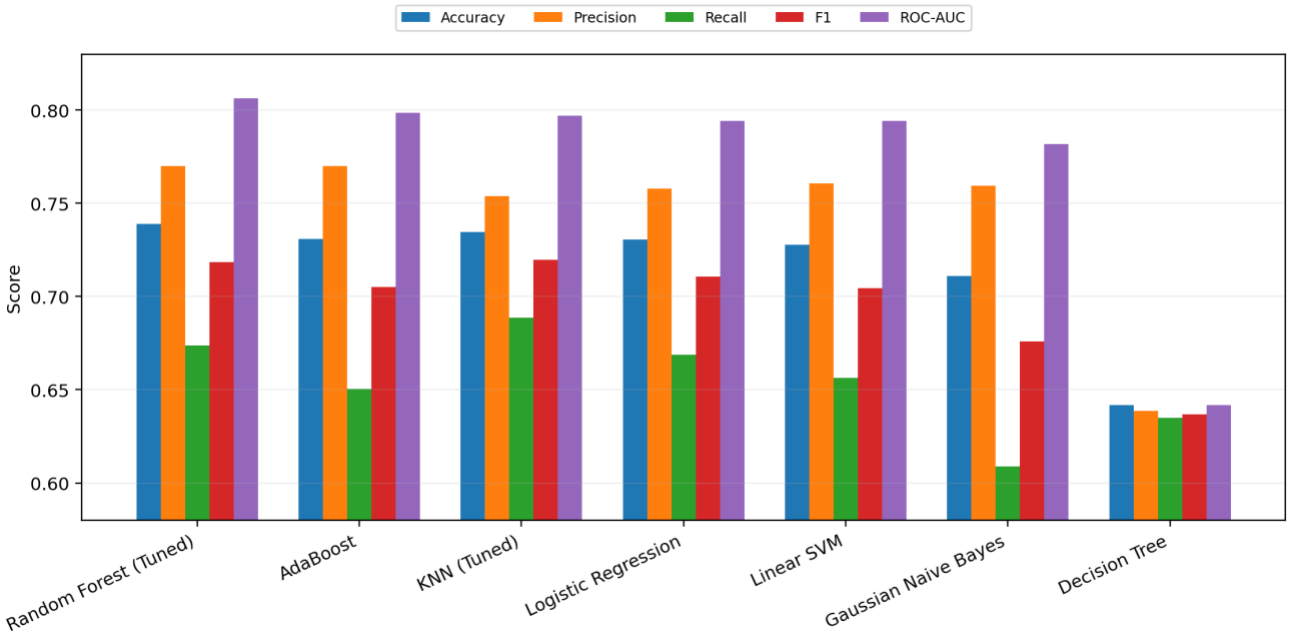


Figure 3.1. Held-out test performance across the seven final classifiers.

Logistic Regression remained competitive despite its simpler form, reaching 0.7305 Accuracy and 0.7943 ROC-AUC. Linear SVM produced very similar ROC-AUC and slightly lower Accuracy.

Gaussian Naive Bayes had lower Accuracy and F1, which is plausible given the strong dependencies among cardiovascular variables and the model's simplifying conditional-independence assumptions. The Decision Tree was clearly the weakest model, with 0.6417 Accuracy and 0.6418 ROC-AUC.

3.3 Overfitting and the Effect of Tuning

The final test table alone does not show how the models arrived at these results. Cross-validation revealed strong overfitting in the default Decision Tree, with training scores far above validation scores. The default Random Forest also fitted the training data very strongly, although its validation and test performance were much better than the single tree. After tuning, the Random Forest used a maximum depth of 10 and minimum leaf size of 8, substantially reducing the train-validation gap. This supports RQ2: tuning improved generalisation behaviour even though the absolute test-score improvement was modest.

Default KNN also showed overfitting relative to its tuned version. Tuning the neighbourhood configuration improved its test Accuracy from approximately 0.6968 to 0.7347 and F1 from approximately 0.6906 to 0.7197. This is a meaningful improvement and shows that KNN is particularly sensitive to configuration. By contrast, the stable performance of Logistic Regression, AdaBoost and Linear SVM indicated less severe train-validation divergence in the experiments conducted.

3.4 Feature Importance and Coefficients

Random Forest feature importance placed systolic blood pressure clearly first, followed by diastolic blood pressure, age, cholesterol and BMI. The stored importance values were approximately 0.4827 for ap_hi, 0.1918 for ap_lo, 0.1351 for age, 0.0917 for cholesterol and 0.0664 for BMI. These values indicate which features the fitted forest used most heavily when reducing impurity across its trees. They should not be interpreted as causal effects or as proof that changing one variable would change cardiovascular outcome.

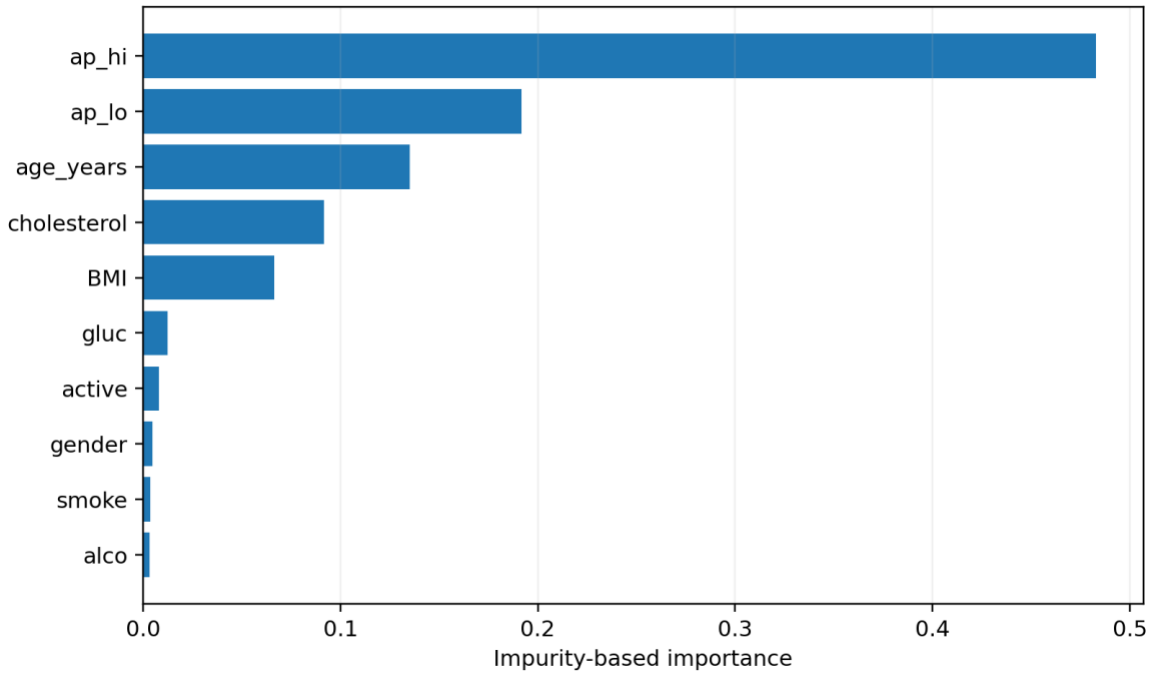


Figure 3.2. Random Forest impurity-based feature importance for the strongest features.

Feature	Random Forest importance
Systolic BP	0.4827
Diastolic BP	0.1918
Age	0.1351
Cholesterol	0.0917
BMI	0.0664

Logistic Regression coefficient inspection also provided directional information, but coefficient magnitudes require caution because numerical variables were standardised whereas categorical/discrete variables were passed through. Cholesterol had a positive coefficient in the fitted model, while physical activity had a negative coefficient. The direction of these coefficients is interpretable within the fitted model, but the difference in feature scaling means raw coefficient magnitudes should not be used as a direct universal ranking of importance.

3.5 Feature-Ablation Experiment

This negative result was useful. It prevented the final methodology from becoming more complicated simply because additional variables were available. It also showed why feature engineering should be evaluated empirically: BMI and pulse pressure are meaningful summaries, but in this dataset they mostly reorganised information that the source measurements already contained.

Feature ablation tested whether the engineered representation materially improved predictive performance. Using Logistic Regression with cross-validation, three representations were compared: an original representation containing height and weight, a representation using BMI, and a representation using BMI plus pulse pressure. The validation results were almost identical. The original representation achieved approximately 0.7275 Accuracy and 0.7905 ROC-AUC; the BMI representation achieved approximately 0.7275 Accuracy and 0.7903 ROC-AUC; and BMI plus pulse pressure also achieved approximately 0.7275 Accuracy and 0.7903 ROC-AUC. This result does not support the hypothesis that adding the engineered variables would materially improve prediction.

Feature representation	Accuracy	Precision	Recall	ROC-AUC
Height + Weight	0.7275	0.7543	0.6661	0.7905
BMI engineered	0.7275	0.7542	0.6662	0.7903
BMI + Pulse Pressure	0.7275	0.7543	0.6662	0.7903

4. Discussion & Critical Evaluation

4.1 Interpreting the Overall Comparison

The strongest models were separated by fractions of a percentage point on some metrics, so the discussion should not exaggerate the ranking. Random Forest was the clearest overall leader because it combined the top Accuracy and ROC-AUC with strong Precision, but KNN produced a slightly better F1 and Recall. AdaBoost produced the highest Precision by only 0.0001 over Random Forest. These values are better interpreted as evidence of different operating behaviour than as large performance gaps.

The closeness of Logistic Regression to the ensemble methods is also important. Logistic Regression reached 0.7305 Accuracy and 0.7943 ROC-AUC, only modestly below Random Forest. This result is consistent with the broader observation from Christodoulou et al. (2019) that more flexible machine-learning methods do not automatically outperform Logistic Regression in clinical prediction studies. In the present dataset, much of the predictive signal may be captured by relatively simple relationships, although the experiment was not designed to prove that explanation.

The relatively close Accuracy values among the stronger models are themselves informative. Random Forest can represent nonlinear interactions and local partitions; Logistic Regression uses a much simpler linear log-odds structure; Linear SVM constructs a linear margin; and KNN uses local distances. Their similar overall Accuracy suggests that a substantial part of the predictive information in this dataset can be captured without an extremely complex model. This is an interpretation rather than a proven property of the underlying population, and it would need external validation or controlled simulation to investigate further.

4.2 Why Random Forest Performed Strongly

One plausible reason is that the dataset contains both continuous measurements and discrete risk indicators, with likely interactions between them. A forest can represent different thresholds and interactions without requiring them to be written down in advance. At the same time, the tuned depth and leaf constraints stop individual trees from becoming as specific as the standalone Decision Tree. Breiman (2001) explains Random Forest performance in terms of the strength of individual trees and the correlation between them; the present results are compatible with that intuition, although the project did not directly estimate tree correlation.

4.3 Why the Single Decision Tree Performed Poorly

The Decision Tree produced the lowest performance and the largest evidence of overfitting. A single tree can continue splitting until it fits fine details of the training sample. Such structure may represent noise or idiosyncratic combinations rather than stable relationships. The contrast with Random Forest is therefore instructive: both are tree-based, but the ensemble aggregates many trees and can control individual tree complexity. The difference between 0.6418 and 0.8062 ROC-AUC cannot be attributed to “trees” as a general algorithmic family; it reflects the difference between one high-variance learner and a regularised ensemble.

4.4 Why Logistic Regression Remained Competitive

Logistic Regression achieved 0.7305 Accuracy and 0.7943 ROC-AUC, only modestly behind the tuned Random Forest. This is important because it demonstrates that increased algorithmic complexity did not automatically produce a large gain. One plausible explanation is that several

influential variables have approximately monotonic relationships with the target, allowing a linear probability model to capture a useful portion of the signal. Another possibility is that the available features do not contain enough information for more flexible models to gain much beyond the baseline. The experiment cannot distinguish these explanations conclusively, but both are testable in future work using interaction analysis, calibration and external datasets.

4.5 Precision-Recall Trade-offs

The additional Random Forest experiment made this trade-off especially clear. The original tuned forest produced Precision 0.7698, Recall 0.6736 and F1 0.7185. A more balanced configuration and controlled threshold produced Precision 0.7334, Recall 0.7420 and F1 0.7377 while Accuracy remained 0.7390 and ROC-AUC 0.8059. The change did not make the classifier universally better; it moved the operating point toward identifying more positives at the cost of more false positives.

This result is why the advanced forest is treated as an additional investigation rather than silently replacing the primary comparison. The original table compares the seven algorithms under their established evaluation setup. The later threshold experiment answers a different question: how much can the Precision-Recall balance change when the decision rule is adjusted using training-data evidence? Keeping those two questions separate makes the dissertation easier to audit.

The main comparison illustrates a practical reason for reporting multiple metrics. Random Forest produced Precision 0.7698 and Recall 0.6736, whereas KNN produced lower Precision, 0.7539, but higher Recall, 0.6885. AdaBoost produced the highest Precision but lower Recall. None of these profiles is automatically preferable without specifying the operational objective. If missing positive cases were treated as more costly, Recall would receive greater emphasis. If false positive predictions were especially costly, Precision might be prioritised. Because this project is a comparative experiment rather than a clinical decision system, the models were not ranked using a medically chosen cost function.

4.6 Engineered Features and Redundancy

The feature-ablation result addresses RQ3. BMI and pulse pressure were reasonable features to investigate because they summarise relationships among source measurements. However, a derived variable cannot automatically add new information when it is a deterministic transformation of variables already available to the model. Pulse pressure is exactly ap_hi minus ap_lo, so including all three creates perfect linear dependence. BMI similarly repackages height and weight. The near-identical ablation results indicate that the predictive information contained in these transformations was already accessible from the original measurements or from the simpler representation.

This result is useful because it prevents a common feature-engineering mistake: assuming that a larger feature set is inherently better. Additional features increase dimensionality and can complicate interpretation, while redundant variables can create multicollinearity in linear models. The decision to retain a compact ten-feature set was therefore based on experimental evidence rather than aesthetic preference.

4.7 Relation to Existing Literature

The project results fit the literature more comfortably when they are treated as dataset-specific evidence. Krittanawong et al. (2020) found wide variation in algorithm performance across

cardiovascular studies, and Bani Hani and Ahmad (2023) likewise reported methodological diversity in ischemic heart-disease prediction. The present results add another example of that variability: Random Forest led overall, but a simple Logistic Regression model remained competitive and KNN led on F1. No claim is therefore made that Random Forest should be preferred in cardiovascular prediction generally.

The result that several conventional algorithms achieved similar overall discrimination is consistent with evidence that more complex machine-learning methods do not automatically outperform simpler statistical models in clinical prediction. Christodoulou et al. (2019) found no performance advantage of machine-learning methods over Logistic Regression in comparisons judged at low risk of bias. In the present experiment, Logistic Regression achieved 0.7943 ROC-AUC, only 0.0119 below the tuned Random Forest, which is compatible with that broader caution against assuming that algorithmic complexity guarantees superior prediction. Krittanawong et al. (2020) and Bani Hani and Ahmad (2023) also report wide variation in algorithms and performance across cardiovascular applications. These studies do not validate the numerical values obtained here because they use different datasets and designs; instead, they provide context for interpreting the close competition among several model families and the importance of validation methodology.

The results also illustrate a reporting issue highlighted by Luo et al. (2016): predictive-model performance is difficult to interpret without details of the data and validation pipeline. For this reason, the filtering rules, final feature representation, train-test split, cross-validation strategy, tuning decisions and held-out evaluation procedure are reported alongside the final performance table.

4.8 Research Questions Revisited

RQ1 asked how the selected algorithms differ under the same experimental conditions. The answer is that Random Forest achieved the strongest Accuracy and ROC-AUC, KNN achieved the strongest Recall and F1 in the primary comparison, AdaBoost achieved the highest Precision by a negligible margin, and Decision Tree performed substantially worse. RQ2 asked whether tuning improves generalisation. The strongest evidence is the reduction in overfitting for Random Forest and the large improvement from default to tuned KNN. RQ3 asked whether engineered features materially improve prediction. The ablation results do not support this hypothesis: alternative feature representations produced almost identical cross-validated performance.

4.9 Limitations of the Study

The most important limitation is that all conclusions are based on one public dataset. A model that performs well here may perform differently on data collected in another country, healthcare system, age distribution or measurement environment. No independent external dataset was used. Therefore the dissertation can compare algorithms within this experiment, but it cannot claim broad clinical generalisability. The Kaggle data card provides a useful variable description but limited detail about the original sampling and measurement process, and the page lists the licence as unknown. These provenance limitations should reduce the strength of any claims extending beyond the dataset itself.

The plausibility thresholds used for height, weight, blood pressure and BMI are researcher decisions. They were deliberately conservative and removed only about 1.96% of observations, but different thresholds could produce slightly different samples and therefore slightly different

model rankings. A stronger future design would conduct sensitivity analysis across multiple reasonable cleaning rules or use a source with documented measurement-quality procedures.

Seven algorithms provide methodological diversity but do not cover all possible supervised learning methods. Gradient boosting libraries such as XGBoost or LightGBM, calibrated models and neural networks were outside the final comparison. Hyperparameter searches were also finite rather than exhaustive. Linear SVM was chosen instead of a nonlinear kernel SVC primarily for computational practicality on the dataset size. These choices are defensible within the project scope but mean that the experiment compares a selected set of algorithms, not the entire space of possible classifiers.

The project aimed for a consistent workflow, but identical preprocessing is not always equivalent to optimal preprocessing. Scaling is important to KNN and SVM but unnecessary for tree-based models. Similarly, the categorical codes for cholesterol and glucose were passed through as ordered numeric values, which assumes meaningful ordering but not one-hot separation. A future comparison could investigate model-specific preprocessing while preserving the same train-test and evaluation protocol.

The final model ranking is based on one held-out test partition, although cross-validation was used during development. The differences among the strongest models are small, and this dissertation does not claim that differences of only a few thousandths are statistically significant. Confidence intervals, bootstrap resampling, repeated cross-validation or nested cross-validation would provide a stronger basis for ranking closely matched algorithms. The evaluation also focused on discrimination and class-prediction metrics rather than probability calibration. ROC-AUC can remain strong even when predicted probabilities are poorly calibrated, so calibration curves or Brier scores would be useful additions if the project were extended toward probability-based decision support. Christodoulou et al. (2019) similarly noted that calibration was frequently omitted in clinical machine-learning studies, which reinforces this as a meaningful limitation rather than a purely technical extra.

Feature importance and coefficients describe the fitted models rather than biological causality. Random Forest impurity importance can favour variables that provide many potential split points, while Logistic Regression coefficient magnitudes depend on feature scaling and coding. The analysis therefore treats these outputs as explanatory aids for the prediction models, not as evidence that a feature causes cardiovascular disease.

These limitations do not make the comparison invalid, but they set clear boundaries around what can be claimed. The results describe the behaviour of these seven algorithms on this cleaned dataset under this particular experimental design. They should not be read as proof that the same ranking would hold for every cardiovascular dataset or every possible preprocessing and tuning strategy.

5. Project Ethics

The project was classified as Ethics Category B0. It used an existing publicly available cardiovascular disease dataset and did not involve recruiting participants, contacting individuals, conducting interventions or collecting new personal information.

The main ethical responsibility was therefore the careful handling and interpretation of secondary data. The project did not attempt to identify individuals, and the dissertation reports only aggregate statistics and model results. Because the data are health-related, the analysis was presented as a machine-learning comparison rather than as medical advice or a clinical decision system.

The experimental work followed the approved scope of the project. Data-quality rules were documented rather than hidden, and model outputs were interpreted cautiously. In particular, feature importance and classifier performance were not presented as evidence of medical causality. This distinction is important because predictive association in a secondary dataset does not establish a clinical relationship.

A further ethical consideration is the risk of over-interpreting performance figures from a health-related dataset. An Accuracy or ROC-AUC value does not by itself demonstrate that a model is suitable for use in healthcare. The dataset has limited information about the original sampling process and no independent clinical validation was carried out. For this reason, the reported results are treated as evidence about algorithm behaviour within this experiment only.

The project also avoided presenting model outputs as recommendations about an individual person. The analysis was designed for experimental comparison rather than diagnosis, treatment advice or individual medical risk assessment. Results were reported at aggregate model level and interpreted only within the scope of the dataset and experimental design. This kept the work aligned with the approved research scope and reduced the risk of the findings being presented as clinical advice.

All reported tables, figures and screenshots use aggregate or project-generated information. No names, contact details or other personally identifying information were collected or introduced during testing. This is consistent with the B0 classification and with the decision not to recruit participants for requirements gathering or evaluation.

6. Conclusion & Future Work

6.1 Conclusion

The three research questions were answered with different kinds of evidence. RQ1 showed that the algorithms were close on overall Accuracy but differed more clearly in error balance and ROC-AUC. RQ2 showed that tuning was valuable where default models overfit, especially Random Forest and KNN. RQ3 was not supported: the engineered representations tested in the ablation study did not materially improve predictive performance. Together, these findings support the main methodological lesson of the project: decisions about preprocessing, validation and measurement are part of the experiment, not background details.

Across the full workflow, seven supervised classifiers were evaluated from the same cleaned dataset using the same held-out test partition and the same set of performance measures. The tuned Random Forest achieved the highest Accuracy and ROC-AUC, while tuned KNN produced the highest Recall and F1 in the primary comparison. The Decision Tree showed the clearest overfitting and the weakest held-out performance.

The main aim was therefore met, but the results also showed why a single ranking is not enough. Model choice changed depending on whether the priority was overall discrimination, positive-prediction reliability or recovery of positive cases. The feature-ablation work added a second lesson: engineered variables should be retained only when they add evidence, not simply because they are easy to construct or intuitively appealing.

6.2 Future Work

The most important future step would be external validation on an independent cardiovascular dataset. This would test whether the relative ranking of algorithms generalises beyond the Kaggle sample. Repeated or nested cross-validation could also provide uncertainty estimates for model differences and reduce dependence on one final test split. More systematic model-specific preprocessing, probability calibration and threshold-selection procedures could be explored. Explainability methods such as permutation importance or SHAP could complement impurity importance, provided their interpretation remains predictive rather than causal. Finally, broader hyperparameter optimisation and additional algorithm families could be investigated, but only within an experimental design that keeps the validation and final-test roles clearly separated.

The project also met the practical objectives set at the beginning. A cleaned and documented dataset was produced, seven classifiers were implemented under a consistent evaluation framework, selected models were tuned using cross-validation, and the final comparison was supported by feature analysis, ablation experiments and generalisation checks. The supporting notebooks and saved result files provide a clear record of how the reported results were produced.

The main value of the project is therefore not a claim that one algorithm should always be selected for cardiovascular data. Instead, it shows how different models can produce similar headline performance while behaving differently on Precision, Recall, F1 and ROC-AUC. It also shows why cleaning choices, leakage prevention, validation design and hyperparameter tuning need to be reported if a model comparison is to be meaningful.

If the work were continued, the first priority would be independent validation rather than adding more models immediately. A result that remains stable on a second dataset would provide much stronger evidence than a small improvement obtained by further tuning on the same source data.

7. BCS Criteria & Self-Reflection

7.1 BCS Project Criteria

Systematic understanding and critical awareness: Addressed through the literature review, comparison of seven supervised learning methods and critical discussion of model behaviour and limitations.

Comprehensive understanding of research techniques: Demonstrated through data cleaning, feature analysis, stratified cross-validation, hyperparameter tuning and multi-metric evaluation.

Originality in the application of knowledge: Demonstrated by applying a consistent comparative framework, investigating redundant engineered features and analysing model-specific trade-offs rather than reporting a single winning score.

Systematic and creative handling of complex issues: Demonstrated through the treatment of abnormal measurements, overfitting, differing preprocessing needs and the interpretation of Precision-Recall trade-offs.

Self-direction and autonomous project work: Demonstrated through the end-to-end implementation of the data pipeline, model experiments, saved outputs and a reproducible notebook workflow.

Critical self-evaluation: Addressed in the reflection below and throughout the limitations section.

7.2 Critical Self-Reflection

One thing I would change is how early I documented decisions. During development, I could remember why I had changed a threshold, feature list or model setting, but those reasons become harder to reconstruct weeks later when writing the dissertation. A more disciplined experiment log would have saved time and made the transition from notebook work to academic writing easier. This became especially obvious when I had to distinguish confirmed hyperparameters from values recreated after a notebook kernel restart.

I also learned that the organisation of the experimental workflow matters as much as the model code itself. Supervisor feedback pushed the project back toward the experimental method, which led me to make the later notebooks more self-contained and to save important outputs explicitly.

Looking back, the most useful change I made during the project was to stop treating the final model score as the main goal. Early on, I was naturally focused on which algorithm would achieve the highest Accuracy. As the experiments developed, it became clear that this was too narrow. The stronger models often had similar Accuracy but noticeably different Precision and Recall. That pushed me to treat the project as a comparison of behaviour rather than a search for one perfect classifier.

The data-cleaning stage also taught me to be more careful about assumptions. The dataset initially looked straightforward because it contained no missing values, but inspection showed extreme blood-pressure, height and weight measurements. Instead of silently removing them, I documented conservative rules and kept track of how many observations were affected. In hindsight, I would plan this validation stage earlier and record every decision from the beginning rather than reconstructing some of the reasoning later.

Another important lesson came from feature engineering. BMI and pulse pressure seemed useful at first because they have an intuitive interpretation, but the ablation results showed almost no predictive gain. This was a useful reminder that a feature should be kept because it adds evidence, not because it sounds meaningful. The same applied to model complexity: Logistic Regression remained competitive with more complex algorithms, while an unconstrained Decision Tree overfit badly.

There were also practical development challenges. Some notebook errors came from inconsistent variable names, missing packages and cells being run in the wrong order. These issues were frustrating at the time, but they improved the way I organised the project. By the end, I was saving cleaned data, result tables and trained models explicitly rather than relying on notebook state.

If I repeated the project, I would define the experiment, file structure and evaluation plan more formally before coding. I would also use model-specific preprocessing from the start, record hyperparameter searches in a single reproducible table, and reserve an independent external dataset for final validation. Overall, the project improved both my technical understanding of supervised learning and my ability to question whether an experimental result is genuinely fair, reproducible and well supported.

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Appendix A. Confirmed Final Model Results

The final primary comparison metrics are reproduced here so that they can be cross-checked against the stored project result files.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7305	0.7579	0.6689	0.7106	0.7943
Decision Tree	0.6417	0.6387	0.6349	0.6368	0.6418
Random Forest (Tuned)	0.7389	0.7698	0.6736	0.7185	0.8062
AdaBoost	0.7309	0.7699	0.6504	0.7051	0.7984
Linear SVM	0.7277	0.7605	0.6563	0.7046	0.7943
KNN (Tuned)	0.7347	0.7539	0.6885	0.7197	0.7970
Gaussian Naive Bayes	0.7111	0.7595	0.6087	0.6758	0.7816

Appendix B. Random Forest Hyperparameters

- n_estimators = 300
- max_depth = 10
- min_samples_split = 5
- min_samples_leaf = 8
- max_features = 'log2'

Appendix C. Supplementary Project Material

The submitted source-code archive should contain the project notebooks, saved result tables and other supporting files required to document the implementation. Large code listings are intentionally not reproduced in full in this dissertation.

The supplementary material should include the original and cleaned datasets used in the project, the notebooks used for preprocessing, analysis and model development, the final comparison CSV files, and the saved model files where available. Keeping these files together makes it possible to trace the values reported in the dissertation back to the experiment outputs.

To reproduce the main comparison, the project should be run using the same final feature set, the fixed 80/20 stratified split with random_state=42, the five-fold Stratified K-Fold configuration and the confirmed Random Forest settings listed in Appendix B. The held-out test partition should remain separate from any additional tuning.

The final submission archive is intended as a record of the implementation rather than a replacement for the dissertation. The dissertation therefore contains the key experimental decisions and results needed to understand the work even if the code is not executed by the marker.